Q6	Verify and document system hardening (audit / checklist)	5 Marks

Objective: Confirm the hardening actually took effect using audit commands, and record it against a checklist.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	TEST	Confirm the system is patched.	`apt list --upgradable`	Empty list
2	TEST	Confirm least-privilege users are in place.	`cat /etc/passwd ; groups`	Expected users only
3	TEST	Confirm the firewall is active with the right rules.	`sudo ufw status verbose`	Active + minimal rules
4	TEST	Confirm only required ports are open.	`nmap -sT -p- 127.0.0.1`	Minimal ports
5	REPORT	Tick each item on the hardening checklist.	Checklist	Completed checklist

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Hardening checklist (lab-vm)

Control	Verified with	Status
Patches applied	apt list --upgradable	**PASS**
Least-privilege users	groups / sudo -l	**PASS**
File permissions	ls -l	**PASS**
Firewall active	ufw status	**PASS**
Minimal open ports	nmap	**PASS**

Every control shows PASS, so the VM meets the baseline hardening standard for this lab.

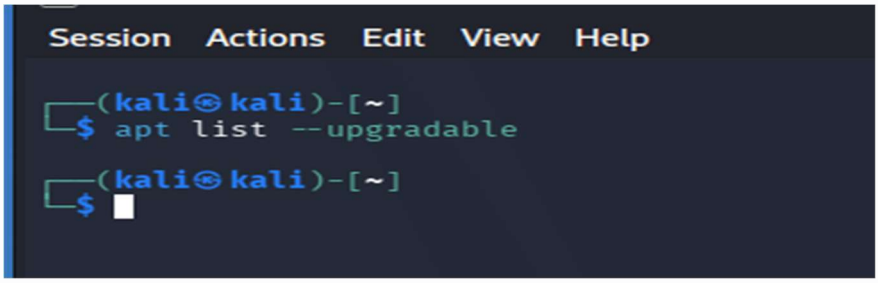
Reusable Template

Hardening checklist template

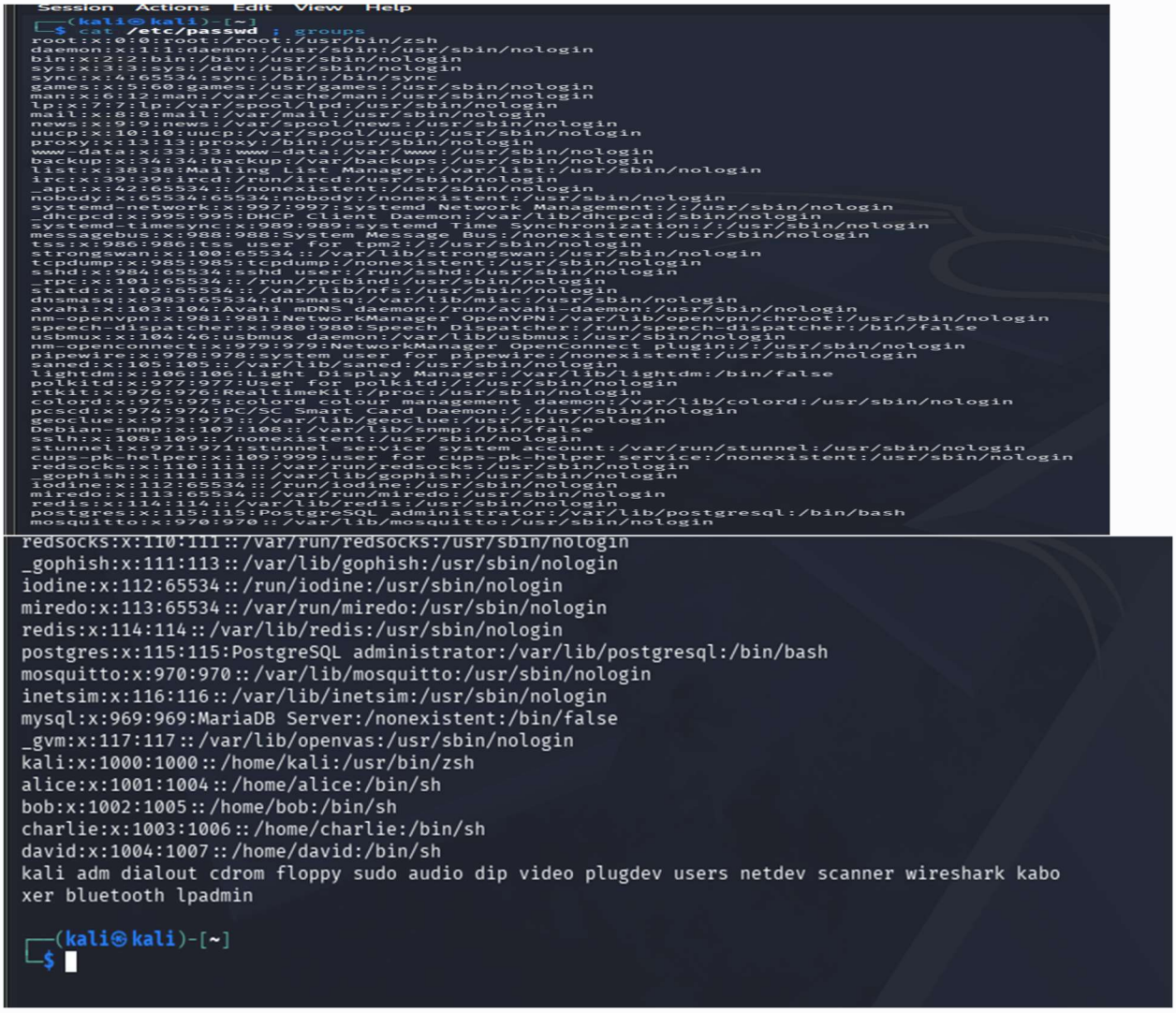
Control	Verified with	Status (PASS/FAIL)
Patches applied	Apt list --upgradable	PASS
Least-privilege users	cat /etc/passwd ; groups /getent group sudo	PASS
File permissions	ls -l	**PASS**
Firewall active	Sudo ufw status verbose	PASS
Minimal open ports	nmap	**PASS**

Your Output / Screenshot (paste result here):

Step 01:



Step 02:



```
┌──(kali㉿kali)-[~]
└─$ getent group sudo
sudo:x:27:kali
```

Step 03:

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

Step 04:

```
┌──(kali㉿kali)-[~]
└─$ nmap -sT -p- 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-03 08:17 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 2.10 seconds
```

Step 05:

```
┌──(kali㉿kali)-[~]
└─$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root   3412 Oct  6 00:22 /etc/passwd
-rw-r----- 1 root shadow 1860 Oct  6 00:34 /etc/shadow
```

End:

The audit commands confirmed that the system is fully patched, only the required users exist, file permissions are restricted, the UFW firewall is active with a default-deny inbound policy, and only the necessary ports are open. Every control on the checklist shows PASS, so the VM meets the baseline hardening standard for this lab.